

A Mini Project Synopsis on

DRIFT DETECTION IN STUDENT WRITTEN ANSWERS

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ABSTRACT

The rapid integration of automated evaluation systems in educational technology has led to widespread adoption of machine learning and deep learning models for assessing student-written answers. These systems rely heavily on statistical patterns and semantic representations learned from historical student responses. However, over time, the characteristics of student answers may change due to variations in curriculum, teaching methodologies, student demographics, exposure to online resources, and generative AI tools. Such changes introduce distributional and semantic shifts in the data, commonly referred to as concept drift. When drift occurs, the performance of automated grading or feedback systems may degrade significantly, leading to inaccurate evaluations and unfair academic assessment.

This project investigates drift detection in student written answers using deep learning-based semantic representations. The primary objective is to design a framework capable of identifying changes in the semantic distribution of student responses over time. Transformer-based language models such as BERT will be utilized to generate contextual embeddings of student answers. These embeddings capture high-level semantic information beyond surface-level textual features. Drift detection techniques, including statistical divergence measures and window-based monitoring algorithms, will be applied to detect deviations between historical and incoming answer distributions.

The proposed approach aims to monitor embedding space shifts and evaluate their impact on automated scoring performance. By identifying drift early, the system can trigger model retraining or adaptation mechanisms, thereby maintaining evaluation reliability. This work contributes to the intersection of educational data mining, natural language processing, and deep learning by addressing the critical challenge of maintaining robust automated assessment systems in dynamic educational environments.

The expected outcome of this project is a systematic methodology for detecting semantic drift in student answers and analyzing its impact on model stability. The findings may enhance fairness, reliability, and long-term sustainability of AI-based academic assessment systems.

INTRODUCTION TO PROBLEM

The rapid digitalization of education has transformed teaching, learning, and assessment. With the growing use of online learning platforms, MOOCs, Learning Management Systems (LMS), and remote examination systems, educational institutions increasingly rely on automated technologies to manage large volumes of student data. One major challenge in this transformation is the automated evaluation of student-written answers. Unlike multiple-choice questions, descriptive answers require semantic understanding, contextual interpretation, and subjective judgment, which traditionally depend on human graders.

Manual grading, although reliable, has several limitations. It is time-consuming, costly, and sometimes inconsistent due to inter-grader variability. As student enrolments increase and digital assessments become more common, scalable and objective evaluation systems have become essential. To address this, machine learning (ML) and deep learning (DL) models have been developed for automated short-answer grading and essay scoring, learning patterns from previously graded responses to approximate human evaluation.

Recent advances in Natural Language Processing (NLP), especially transformer-based models such as BERT, RoBERTa, and GPT-style architectures, have improved automated evaluation systems. These models generate contextual embeddings that capture semantic relationships within text, enabling better understanding of meaning, sentence structure, and linguistic patterns compared to traditional feature-based approaches.

However, most deep learning models assume that the statistical properties of training data remain stable over time. In real-world educational environments, student-written answers continuously evolve due to changes in curriculum, teaching methods, study resources, and technological tools. Different student cohorts may also display varying writing styles, vocabulary usage, and conceptual interpretations.

These evolving patterns can change the distribution of student responses. When this happens, models trained on historical data may struggle to generalize to new answers because the relationship between inputs (student responses) and outputs (grades) changes. This phenomenon is known as concept drift.

Concept drift is a major challenge for ML systems used over long periods. In automated grading systems, it can lead to inaccurate scoring, biased feedback, and unreliable evaluations. While much research has focused on improving grading accuracy, relatively little attention has been given to monitoring system performance after deployment. Without drift detection mechanisms, automated grading systems risk becoming outdated and unreliable.

This project addresses this issue by investigating drift detection in student-written answers using deep learning-based semantic representations. By analysing distributional shifts in embeddings generated by transformer models and applying statistical drift detection techniques, the framework aims to identify changes in student response patterns.

Early detection of drift can enable proactive model maintenance strategies such as retraining or fine-tuning, ensuring automated grading systems remain reliable and aligned with evolving educational practices. This work contributes to developing sustainable and trustworthy AI systems for education.

PROBLEM STATEMENT

Automated grading systems based on deep learning models are increasingly used to evaluate student-written answers in online and hybrid educational environments. These systems are typically trained on historical datasets containing previously graded responses. Once deployed, they are expected to evaluate future student answers accurately and consistently. However, most existing systems assume that the statistical and semantic characteristics of student responses remain stable over time. In practical educational settings, this assumption is unrealistic.

Student-written answers evolved due to changes in curriculum content, teaching methods, assessment patterns, access to digital resources, and differences across student cohorts. Additionally, the growing use of AI-assisted writing tools can alter the linguistic and structural properties of responses. Such changes may lead to shifts in vocabulary usage, semantic emphasis, conceptual explanation styles, and answer length distributions. As a result, the underlying data distribution encountered during deployment may differ significantly from the training data distribution.

When this distributional shift occurs, the performance of the automated grading model can degrade. The model may misinterpret new writing patterns, assign inaccurate scores, or fail to generalize to unseen semantic structures. This degradation not only reduces system reliability but may also introduce unfairness and bias in academic evaluation. In high-stakes educational contexts, such inconsistencies can have serious consequences.

This phenomenon is formally known as concept drift, where the relationship between input features (student answers) and output labels (scores) changes over time. In deep learning-based systems, drift may manifest as shifts in high-dimensional embedding space representations generated by transformer models. However, most current automated grading frameworks lack mechanisms to monitor and detect such drift after deployment. They operate as static models without continuous evaluation of data stability.

Therefore, the core problem addressed in this project is twofold:

1. How to effectively detect semantic and distributional drift in student-written answers using deep learning-based representations.
2. How to design an adaptive retraining strategy that updates the grading model when significant drift is detected, ensuring sustained performance and fairness.

The project aims to develop a complete pipeline that includes semantic embedding extraction using transformer models, statistical drift detection mechanisms to monitor embedding distribution changes, performance evaluation monitoring, and a retraining or fine-tuning module triggered by detected drift. By integrating detection and adaptation into a unified framework, the system can maintain robustness in dynamic educational environments.

In summary, the central challenge is not only building an accurate grading model but ensuring its long-term reliability in the presence of evolving student writing patterns. Addressing this challenge requires combining deep learning, statistical monitoring, and adaptive model maintenance strategies into a cohesive and practical solution.

OBJECTIVES

The primary objective of this project is to design and develop a deep learning-based framework capable of detecting semantic drift in student-written answers and adapting the grading model to maintain consistent performance over time. The study aims to move beyond static automated grading systems by incorporating continuous monitoring and adaptive learning mechanisms.

To achieve this, the specific objectives of the study are as follows:

1. To Develop a Baseline Deep Learning Model for Automated Answer Evaluation

A transformer-based model such as BERT will be fine-tuned on a labelled dataset of student responses to create a baseline system for automated answer evaluation. This model will serve as the reference system whose performance will be monitored over time.

2. To Generate High-Dimensional Semantic Representations of Student Answers

Contextual embeddings will be extracted from transformer models to capture semantic meaning and contextual relationships within student answers. These embeddings will be used to monitor changes in answer patterns.

3. To Detect Distributional and Concept Drift in Student Answers

A key objective is to identify statistical and semantic shifts between historical (training) data and incoming student responses. Drift detection techniques such as:

- Kullback-Leibler (KL) Divergence
- Jensen-Shannon Divergence
- Maximum Mean Discrepancy (MMD)
- Window-based monitoring methods (e.g., ADWIN)

will be explored to measure changes in embedding distributions. The system will determine whether detected shifts are statistically significant enough to impact model performance.

4. To Monitor Model Performance Degradation

Performance metrics such as accuracy, F1-score, correlation with human grades, or mean squared error will be tracked to determine whether detected drift affects grading performance.

5. To Design an Adaptive Retraining Strategy

When significant drift is detected, the system will trigger retraining strategies such as incremental learning, periodic retraining with updated datasets, or fine-tuning the transformer model.

6. To Evaluate the Effectiveness of the Complete Drift Detection Pipeline

Finally, the project aims to assess whether the proposed pipeline successfully identifies drift early and restores performance through retraining. Comparative analysis will be conducted between:

- Static (non-adaptive) grading models
- Drift-aware adaptive models

This evaluation will demonstrate the practical importance of incorporating drift detection mechanisms in real-world educational AI systems.

LITERATURE REVIEW

The field of automated assessment of student-written answers has evolved significantly over the past two decades. Early research primarily focused on rule-based and statistical methods, while recent developments have shifted toward deep learning-based semantic models. However, the challenge of maintaining long-term model reliability through drift detection remains relatively underexplored.

1. Automated Essay Scoring and Short Answer Grading

Early automated grading systems used handcrafted features such as word frequency, sentence length, grammar rules, and keyword matching. Systems like Project Essay Grade (PEG) and e-rater combined these linguistic features with regression or classification models to predict scores [1]. Later approaches introduced machine learning methods such as Support Vector Machines (SVM), Random Forests, and TF-IDF representations, improving performance but still relying on manual feature engineering [2]. The adoption of deep learning models like CNNs and LSTMs further improved automated grading by learning textual patterns directly from data [3].

2. Transformer-Based Models in Educational NLP

Transformer architectures, particularly BERT (Bidirectional Encoder Representations from Transformers), significantly improved Natural Language Processing by enabling contextual understanding of text through self-attention mechanisms [4]. These models have been applied to automated essay scoring and short-answer grading tasks, generating contextual embeddings that capture semantic and syntactic relationships [5], [6]. However, most transformer-based grading systems are trained and evaluated on fixed datasets and rarely consider changes in student writing patterns over time.

3. Concept Drift in Machine Learning Systems

Concept drift refers to changes in data distribution or the relationship between inputs and outputs over time [7]. It can occur as sudden, gradual, incremental, or recurring drift. Several techniques have been proposed to detect such changes, including statistical divergence measures like Kullback-Leibler divergence and Jensen-Shannon divergence, window-based algorithms such as ADWIN, and distribution comparison methods like Maximum Mean Discrepancy (MMD) [8], [9]. However, most drift detection research focuses on structured numerical data rather than textual embeddings.

4. Drift Detection in Natural Language Processing

Drift detection in NLP is an emerging area of research. Language models deployed in real-world environments may experience performance degradation due to vocabulary evolution, domain shifts, or topic changes [10]. Monitoring embedding distributions and applying domain adaptation or continual learning techniques have been proposed to address this issue [11]. However, these methods are rarely applied in automated educational assessment systems.

5. Research Gap

Although deep learning has improved automated grading systems, limited work has addressed their long-term stability. Most models assume static data distributions and lack mechanisms to detect semantic changes in student responses. There is also limited research integrating embedding-based drift detection with adaptive retraining in educational contexts. This study aims to address this gap by proposing a deep learning-based drift detection and adaptive retraining framework for student-written answers.

METHODOLOGY

The proposed methodology introduces a drift-aware automated grading framework that integrates deep learning-based semantic representations, statistical drift detection, and adaptive retraining to maintain reliable automated evaluation.

1. Dataset and Data Organization

The framework uses labelled datasets containing student-written answers and corresponding human-assigned scores, such as the Automated Student Assessment Prize dataset or similar institutional datasets. The data is divided into historical training data, validation data, and incoming batches that simulate real-world deployment. Temporal segmentation is used so that later batches reflect evolving student writing patterns.

2. Data Preprocessing

Student responses are preprocessed by cleaning text, tokenizing using the transformer tokenizer, padding or truncating sequences, and generating attention masks. Transformer models automatically capture contextual relationships, reducing the need for manual feature engineering.

3. Semantic Embedding Extraction Using Transformer Models

A pre-trained transformer model such as BERT generates contextual embeddings for student answers. The pooled output or classification token embedding is extracted, forming a high-dimensional semantic vector representing meaning and context. These embeddings are used both for grading and drift detection.

4. Baseline Automated Grading Model

The grading model is developed by fine-tuning the transformer on labelled responses. The task may be framed as regression for continuous scores or classification for grade categories. The final layer is replaced with a task-specific fully connected layer and trained using loss functions such as Mean Squared Error or Cross-Entropy Loss. Performance is measured using metrics like accuracy, F1-score, Quadratic Weighted Kappa, and correlation with human grades.

5. Drift Detection Mechanism

Changes in semantic embeddings of incoming responses are monitored using statistical measures such as Kullback-Leibler Divergence, Jensen-Shannon Divergence, and Maximum Mean Discrepancy. Window-based techniques like Adaptive Windowing (ADWIN) help detect distributional shifts in streaming or batch data.

6. Performance Monitoring

Model performance is continuously evaluated using validation or newly labelled data. Metrics such as accuracy, F1-score, and Quadratic Weighted Kappa help identify performance degradation caused by drift.

7. Adaptive Retraining Strategy

When significant drift is detected, the system performs adaptive retraining through incremental fine-tuning, periodic retraining with updated datasets, or sliding-window training where older data is gradually replaced.

8. Overall Pipeline Operation

Student answers are preprocessed, embeddings are generated using a transformer model, and scores are predicted. Embedding distributions and performance metrics are monitored continuously. If drift is detected, retraining is triggered to maintain grading accuracy and reliability.

EXPECTED OUTCOME

The proposed drift-aware grading framework is expected to demonstrate significant improvements in long-term stability, reliability, and fairness of automated evaluation systems for student-written answers. Unlike traditional static grading models, the proposed system continuously monitors semantic shifts and adapts accordingly, which should lead to measurable performance benefits.

First, the baseline transformer-based grading model is expected to achieve strong initial performance in predicting student scores. Due to the use of contextual embeddings generated by models such as BERT, the system should demonstrate high semantic understanding and strong correlation with human-assigned grades. Performance metrics such as accuracy, F1-score, and Quadratic Weighted Kappa are expected to indicate reliable grading quality under stable data conditions.

However, when exposed to temporally separated or evolving student responses, the static baseline model is expected to exhibit gradual performance degradation. This degradation may occur due to changes in vocabulary usage, answer structure, conceptual explanation styles, or writing patterns. Such changes will likely cause measurable shifts in embedding distributions and a decline in grading performance metrics. This behavior will validate the presence of concept drift in student-written answers.

The drift detection module is expected to successfully identify statistically significant changes in embedding space using divergence measures such as Kullback-Leibler Divergence, Jensen-Shannon Divergence, and Maximum Mean Discrepancy. The system should detect drift before substantial performance collapse occurs, enabling early intervention. Window-based monitoring techniques are expected to improve detection robustness and reduce false positive alerts.

Upon activation of the adaptive retraining mechanism, the grading model is expected to recover lost performance. Fine-tuning the transformer model on newly labeled responses should allow it to adapt to updated semantic patterns and restore evaluation accuracy. Compared to the static baseline model, the adaptive system is expected to demonstrate improved stability across different data batches and maintain consistent grading quality over time.

Additionally, the proposed framework is expected to enhance fairness in automated assessment. By identifying and correcting performance degradation caused by distributional shifts, the system reduces the risk of biased or inconsistent grading for newer student cohorts. This is particularly important in dynamic educational environments where student demographics, curriculum structure, and technological exposure continuously evolve.

Overall, the primary expected outcome is the validation of a complete drift-aware automated grading pipeline that combines deep learning-based semantic modeling with statistical monitoring and adaptive retraining. The results are expected to demonstrate that incorporating drift detection mechanisms significantly improves the long-term robustness of AI-driven educational assessment systems compared to static models.

CONCLUSION

This study presents a drift-aware deep learning framework for detecting and adapting to semantic changes in student-written answers. While transformer-based models such as BERT have significantly improved automated grading accuracy, most existing systems assume that student response distributions remain stable over time. In real educational environments, this assumption does not hold due to evolving curricula, changing student cohorts, and shifts in writing patterns. As a result, static grading models risk performance degradation and reduce reliability.

The proposed framework addresses this limitation by integrating semantic embedding extraction, statistical drift detection, performance monitoring, and adaptive retraining into a unified pipeline. By continuously analyzing changes in embedding space and monitoring evaluation metrics, the system can identify concept drift before it severely impacts grading quality. The inclusion of adaptive retraining mechanisms ensures that the model remains aligned with current student writing patterns and maintains consistent evaluation standards.

The key contribution of this work lies in shifting the focus from building high-accuracy grading models to ensuring their long-term robustness and fairness. Instead of treating automated grading as a one-time training problem, this study emphasizes continuous monitoring and adaptation in dynamic educational settings. Such an approach is essential for deploying reliable AI systems in real-world academic environments.

In conclusion, incorporating drift detection and adaptive learning strategies into automated assessment systems enhances their stability, fairness, and sustainability. This framework contributes to the advancement of trustworthy deep learning applications in education and highlights the importance of maintaining model performance over time rather than solely optimizing initial accuracy.

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